

Quasiregular Reeb Dynamics on Pairwise-Coprime Brieskorn Three-Manifolds

HCS-C370 theorem package

4 September 2026

Abstract

For every pair of odd coprime integers $3 \leq p < q$, we close the normalized Reeb dynamics of the Brieskorn link $\Sigma(2, p, q)$. There are exactly three exceptional simple orbit circles, of periods $2p$, $2q$, pq , and a principal period $2pq$. A divisibility rule gives every fixed-time component and proves it is Morse–Bott. Missing-coordinate trivializations give the transverse rotations, return determinants, first degeneracies, and all pre-degeneracy Conley–Zehnder indices. The orbit space is $S^2(2, p, q)$, and a declared Milnor-fiber capping convention gives $\mu_{\text{RS}} = 2(2pq)\chi_{\text{orb}} = -2pq + 4p + 4q$; this is positive only for $(p, q) = (3, 5)$ and never zero. Exhaustive exact computation is a regression receipt, not the proof.

1 Frozen contact system and main result

Put $a = (a_0, a_1, a_2) = (2, p, q)$ and

$$f(z) = z_0^2 + z_1^p + z_2^q, \quad \Sigma_{p,q} = f^{-1}(0) \cap S^5 \subset \mathbb{C}^3.$$

The origin is the only critical point of f , so $\Sigma_{p,q}$ is a smooth closed three-manifold. We freeze the scale

$$\alpha = \frac{i}{4\pi} \sum_{j=0}^2 a_j (z_j d\bar{z}_j - \bar{z}_j dz_j).$$

This is the standard weighted Brieskorn contact form in the normalization used throughout this paper.

Theorem 1 (Complete Reeb atlas) *The Reeb vector field and its flow are*

$$R = 2\pi i \sum_{j=0}^2 \frac{1}{a_j} (z_j \partial_{z_j} - \bar{z}_j \partial_{\bar{z}_j}), \quad \Phi_t(z_j) = e^{2\pi i t/a_j} z_j.$$

There are three exceptional simple circles, with support and primitive period

$$\{0, 1\} : 2p, \quad \{0, 2\} : 2q, \quad \{1, 2\} : pq,$$

whereas the principal period is $d = 2pq$. For real T , let $J_T = \{j : T/a_j \in \mathbb{Z}\}$. The fixed set is empty for $|J_T| < 2$, the corresponding exceptional circle for $|J_T| = 2$, and all of $\Sigma_{p,q}$ for $|J_T| = 3$. Every nonempty component is Morse–Bott.

In the ambient missing-coordinate complex-line trivializations, the three transverse rotation numbers are

$$\rho_{01} = \frac{2p}{q}, \quad \rho_{02} = \frac{2q}{p}, \quad \rho_{12} = \frac{pq}{2}.$$

Thus $\det_{\mathbb{R}}(I - P) = 4 \sin^2(\pi\rho)$, and first degeneracy occurs at covers $q, p, 2$. Before that cover,

$$\mu_{\text{CZ}}(\gamma^r) = 2[r\rho] + 1.$$

The common-period orbit space is $S^2(2, p, q)$, with

$$\chi_{\text{orb}} = \frac{1}{2} + \frac{1}{p} + \frac{1}{q} - 1.$$

In the standard Milnor-fiber capping trivialization, the principal family has

$$\mu_{\text{RS}} = 2d\chi_{\text{orb}} = -2pq + 4p + 4q.$$

Both quantities are positive exactly at $(p, q) = (3, 5)$, negative otherwise, and never zero.

2 Normalization and primitive orbit types

The weighted ambient two-form is positive, and its contact-type restriction to the link is $d\alpha$. Direct evaluation gives, coordinate by coordinate,

$$\alpha(R) = \sum_j |z_j|^2 = 1, \quad \iota_R d\alpha = -d\|z\|^2, \quad df(R) = 2\pi i f.$$

Therefore R is tangent to both defining hypersurfaces, and $\iota_R d\alpha$ vanishes on the tangent of the link. This proves the stated normalization rather than fixing it only up to a positive constant.

A link point cannot have only one nonzero coordinate. On a two-coordinate locus of weights a_i, a_j , the moduli are the unique positive solution of

$$r^{a_i} = s^{a_j}, \quad r^2 + s^2 = 1.$$

The phase equation cuts out a connected one-dimensional subtorus because $\gcd(a_i, a_j) = 1$. Hence each such locus is one embedded circle and one Reeb orbit. A point of support S returns first at $\text{lcm}(a_j : j \in S)$. Pairwise coprimality of $2, p, q$ now gives

support	01	02	12	012
period	$2p$	$2q$	pq	$2pq$
isotropy order	q	p	2	1
orbit-quotient dimension	0	0	0	2

The final entry is the structural obstruction to treating all primitive orbits as a discrete ledger.